

WSA (Within Shift Analysis)

Updated Project Overview & Current Web Application Enhancements

Project purpose

Quality inspection, audit management, NOK case tracking, analytics and reporting for manufacturing environments.

Current update scope

The web application has been visually and functionally improved across the Home, Dashboard, Records, Observations, Analytics, and Reports & Export modules.

Updated date

21 September 2026

Update note

This document keeps the original project scope and technology stack, while documenting the web-page changes made in the current working copy. Core authentication, records, NOK cases, CSV export, and the SQLite-based architecture remain part of the application.

1. Project Overview

WSA (Within Shift Analysis) is a quality inspection and audit management system developed to manage production audit records, Non-OK (NOK) cases, analytics, observations, and reporting within manufacturing or industrial environments. The system is intended for Auditors, Supervisors / Admins, and Quality Management Teams.

The application supports centralized quality monitoring, defect tracking, image evidence capture, data visualization, CSV reporting, and role-based access control.

Key Features

- Secure login and session-based authentication.
- Separate Admin and User roles with role-specific record and case permissions.
- Production audit record creation, editing, deletion, and review.
- NOK case management with defect details and up to three images per case.
- Dashboard and analytics views for quality and production monitoring.
- Filtered CSV export for reporting, documentation, external analysis, and audit review.

Authentication & Authorization

- Admins can register and login, add/edit/delete records, manage NOK cases, view dashboards and analytics, and download filtered reports.
- Users can login and create records, add NOK details, upload images, view analytics, and edit only their own records/cases.
- Users cannot delete records or edit records created by other users.

Current UI change

The application now uses a consistent WSA layout with a structured left navigation, branded header, active-page highlighting, system status indicator, and a dedicated logout action.

2. Record & NOK Case Management

Record Management

Inspection records contain date, auditor name, type number, parts checked, NOK count, shift, and line information. Based on the NOK count, the system automatically generates the required number of NOK cases.

Record field	Purpose
Date	Identifies when the audit was performed.
Auditor Name	Identifies the person performing the audit.
Type Number	Identifies the inspected product/type.
Parts Checked	Records production volume inspected.
NOK Count	Records the number of rejected/non-OK units.
Shift	Supports within-shift analysis.
Line Information	Supports line-wise quality analysis.

NOK Case Management

Each NOK case can contain rejection status, bin number, particle location, body details, valve details, nozzle details, magnet details, remarks, and up to three images per case.

- NOK cases provide case-level defect tracking instead of only a summary count.
- Image evidence can be attached to support inspection review and root-cause analysis.
- The Observations module provides a dedicated view of these cases.

Current UI change

- The Records page now includes summary KPI cards for Total Records, OK Records, NOK Records, and Parts Checked.
- The Records table uses clear OK/NOK status badges, shift badges, record-code badges, and role-aware Edit/Delete actions.
- Client-side search was added so users can quickly search by record code, auditor, type, shift, line, or other visible record text.

3. Dashboard & Analytics - Updated

The original dashboard scope covers daily, weekly, monthly, auditor-wise, and shift-wise analysis, with filters for shift, line, auditor, type, and date range. Visualized charts are used to analyze production quality, rejection trends, auditor performance, and shift performance.

Current Dashboard enhancements

- Four KPI cards: Total Audits, Parts Checked, NOK Cases, and Rejection Rate.
- Centralized filters for From Date, To Date, Shift, Line, Auditor, and Type.
- Daily Audit vs NOK trend visualization.
- Shift Performance visualization.
- Line-wise NOK visualization.
- Recent NOK Cases panel with record, date, type, shift, and line context.
- Auditor day-wise record summary.

Current Analytics enhancements

- Analysis selector for Shift, Line, Auditor, and Type analysis.
- Date, shift, and line filters for focused analysis.
- Separate production-volume chart for Parts Checked by Group.
- Separate quality chart for NOK Rate by Group so rejection rate remains readable even when production volumes are much larger.
- Analysis Breakdown table showing group, parts checked, NOK, and NOK rate.

Quality-analysis design change

The earlier single-scale chart compared large production counts with smaller NOK counts. The updated Analytics page separates Parts Checked from NOK Rate, making the quality metric easier to interpret.

4. Observations & Reporting - Updated

Observations Module

The observations section allows users to view NOK case details, apply filters, analyze defect observations, and review rejection information.

Current Observations enhancements

- Summary cards for NOK Cases, Image Evidence, Morning Cases, and Evening/Night Cases.
- Search field for case code, record code, type, shift, line, rejection station, and particle location.
- Date, shift, and line filters.
- Case review table with case code, record code, date, shift, type, line, rejection information, particle location, evidence, and actions.
- Evidence buttons for available inspection images.
- Role-aware Edit/Delete controls.

Download & Reporting

The system supports CSV export for record data only, combined record + NOK case data, and filtered data export. This supports reporting, documentation, external analysis, and audit review.

Current Reports & Export enhancements

- Dedicated Reports & Export page with clear distinction between Record Data and Record + NOK Case Data.
- From Date and To Date filters retained for export configuration.
- Explicit Include Case Details option for combined export.
- Reporting workflow presentation: Filter -> Select Detail -> Download -> Review.

5. Current Web Page Changes Implemented

Module	Current change
Home	Redesigned landing page with WSA welcome section, quick-action cards, user information, and audit workflow overview.
Global Layout	Consistent Bosch/WSA header, left sidebar navigation, active navigation state, system-active indicator, and user profile.
Dashboard	KPI cards, centralized filters, trend/distribution charts, recent NOK cases, and auditor activity summary.
Records	Summary metrics, filters, client-side search, OK/NOK badges, cleaner table, and role-aware actions.
Observations	NOK case summary, search/filter controls, defect-oriented columns, evidence buttons, and case actions.
Analytics	Separate Parts Checked and NOK Rate visualizations plus an analysis breakdown table.
Reports & Export	Card-based explanation of export modes, date range selection, case-detail option, and reporting workflow.
Navigation	Consistent active-page highlighting and clearer terminology such as Reports & Export.
Logout	Distinct exit action styling to separate it from normal navigation.

Demonstration flow

A clear end-to-end demo flow is now available: Login -> Home -> Dashboard -> Filters -> View Records -> Observations -> Analytics -> Reports & Export -> Logout.

Design approach

- The core FastAPI/SQLite project structure remains unchanged in principle.
- The main visual work is concentrated in Jinja templates, CSS, and browser-side JavaScript.
- Dashboard and analytics calculations use application data rather than hard-coded demo values in the UI.
- The updated UI is designed for clear manufacturing-quality review rather than decorative styling.

6. Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Python, FastAPI
Database	SQLite (future migration planned to PostgreSQL)
ORM & Security	SQLAlchemy, Passlib
Template Engine	Jinja2 Templates
Validation	Pydantic

Key Python libraries / framework usage

FastAPI backend imports include FastAPI, Request, Depends, Form, HTTPException, HTMLResponse, RedirectResponse, StreamingResponse, Jinja2Templates, and StaticFiles. Database and ORM functions use SQLAlchemy engine/session/model components. Authentication uses Passlib CryptContext, and validation uses Pydantic BaseModel.

Updated Project Structure

```
WSA/  
backend/ -> auth.py, database.py, main.py, models.py, schemas.py  
static/ -> style.css, script.js, analytics.js, dashboard.js, records.js, uploads/  
templates/ -> Jinja2 web templates  
requirements.txt  
README.md
```

Current working copy additions / updates are primarily within the template and static front-end files, with supporting route logic in main.py for dashboard/analytics/observation handling.

7. How to Run the Project

The original run sequence remains:

1. Clone the repository

```
git clone  
cd WSA
```

2. Create a virtual environment

```
python -m venv venv  
3. Activate the environment - Windows: venv\Scripts\activate
```

4. Install dependencies

```
pip install -r requirements.txt
```

5. Run the application

```
uvicorn backend.main:app --reload
```

6. Open in browser

```
http://127.0.0.1:8000
```

8. Future Scope

- Migration from SQLite to PostgreSQL.
- Cloud deployment support.
- Email notification system.
- Role-based advanced permissions.
- Real-time analytics dashboard.
- PDF report generation.
- AI-based defect analysis.
- Mobile responsive enhancements.
- API integration with ERP/MES systems.
- Multi-plant support.
- Data backup and recovery.
- Audit history tracking.

9. Benefits

- Centralized quality monitoring.
- Better production tracking.
- Improved defect analysis.
- Faster report generation.
- Enhanced audit management.
- Easy data visualization.
- Better accountability through role management.

10. Conclusion

WSA (Within Shift Analysis) is a complete audit and production quality management system developed using FastAPI, Python, SQLAlchemy, HTML, CSS, JavaScript, and Bootstrap. The current working copy adds a more structured enterprise-style interface, clearer analytics, better case visibility, and a more polished reporting workflow while retaining the original application purpose and architecture.

Current status

The current web application has been updated across the major user-facing modules and is suitable for a guided demonstration of the improved WSA interface and workflow.